
Building Signals through Fourier Coefficients

Overview of the Lab: In this lab we are going to create and synthesize periodic signals. When we compared signals last week. This lab works on building the notation of the Fourier Series, and computer skills for plotting, working with vectors, and for-loops. Note in the Lab how this works can translate to data compression applications.

Problem Statement Let's pose the following scenario. You are on a remote island working in a secret lab, where on the mainland they are transmitting you a "compressed" sequence of time series data, $X(t)_{sum}$. You know the system is being sampled at 80Hz with a fundamental frequency $f_0 = 4Hz$. The mainland's transmissions are a set of complex numbers, A_k , where k is linked to the k^{th} Fourier coefficient. Assume that any other not listed coefficients are zero (i.e. $0 + 0i$). You will have to simulate (synthesize) the sequence for 1 second in the time domain.

$$a_{-9} = 3 - 2.5i,$$

$$a_{-7} = 2 - 2.5i,$$

$$a_{-4} = 3 - 3i,$$

$$a_{-2} = 8 - 5i,$$

$$a_0 = 5 + 0i,$$

$$a_2 = 8 + 5i,$$

$$a_4 = 3 + 3i,$$

$$a_7 = 2 + 2.5i,$$

$$a_9 = 3 + 2.5i,$$

- Calculate the following phasors using Matlab in polar form as $Ae^{j\phi} \cdot e^{j2\pi ft}$ for each set of Fourier coefficients. The code should use a for-loop where the amplitude and phase are held in the vectors, *Amp* and *Phi* (provide the code and solution).

$$\mathbf{Amp} = [A_{-9}, A_{-8}, A_{-7}, A_{-6}, A_{-5}, A_{-4}, A_{-3}, A_{-2}, A_{-1}, A_0, A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9]$$

$$\mathbf{Phi} = [\phi_{-9}, \phi_{-8}, \phi_{-7}, \phi_{-6}, \phi_{-5}, \phi_{-4}, \phi_{-3}, \phi_{-2}, \phi_{-1}, \phi_0, \phi_1, \phi_2, \phi_3, \phi_4, \phi_5, \phi_6, \phi_7, \phi_8, \phi_9]$$

- In order to construct these components of the $X(t)_{sum}$, we first must make a time vector t . Provide the MatLab code for the **time vector**, "t" that will be used in the Fourier synthesis equation. This will be held as a vector and used to build your sinusoid components.
- Create another four loop that will loop through the a_k fourier coefficients which will create each sinusoidal component $X_k(t)$. These sinusoidal components will be held in a Matrix, $M(K, N)$, where k is the number of Fourier coefficients and N is the length of the signal. Recall inverse Eulers Formula which was discussed in class.

$$A \cos(\overbrace{\omega_o t + \phi}^{\theta}) = \frac{A}{2} e^{j\phi} e^{j\omega_o t} + \frac{A}{2} e^{-j\phi} e^{-j\omega_o t}$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

- Provide the code for how the Fourier synthesis will be implemented.

Fourier Synthesis:

A signal $X(t)_{sum}$ is generated by,

$$X(t)_{sum} = a_o + \sum_{i=1}^N a_i \cos(2\pi f_i t + \phi_i), \quad (1)$$

such that $f_i = K_i f_o$, where $K_1, K_2, \dots, K_n$ are positive integers and $\gcd(K_1, K_2, \dots, K_N) = 1$. Then $X(t)$ is periodic with $T_0 = \frac{1}{f_o}$, where $f_o \triangleq$ fundamental frequency.

- Plot 3 rows of the Matrix that are tied to an a_k . Describe the signals for each plot and its relationship to the components of a sinusoidal wave.
- Add the Matrix rows to then construct the $X(t)_{sum}$ signal, using the matlab function "sum".
- What is the DC offset of the synthesized signal and how does this relate to the fourier coefficients?
- Provide a plot of the synthesized signal. Use the grid on feature on the plot and describe what the graph represents.